The Transdimensional Energy Descent Cosmological Program: A Unified Cosmological Framework Based on Dimensional Interface Dynamics

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Abstract

Mainstream cosmology and particle physics currently adopt a domain-separated effective theoretical framework. While the ΛCDM model successfully describes the large-scale evolution of the universe, the physical nature of dark matter and dark energy remains unclear, and the underlying mechanisms of singularities and early cosmic inflation are yet to be identified. The Standard Model of particle physics unifies three fundamental microscopic interactions but cannot incorporate gravity; the ontodynamic mechanism of mass-energy conversion has also long remained unexplained. General relativity and quantum mechanics present irreconcilable contradictions at the Planck scale. Leading quantum gravity proposals rely on numerous additional ad hoc assumptions and have long lacked directly testable observational predictions.

This paper proposes a unified cosmological research program built on transdimensional descent of higher-dimensional energy and dimensional interface tension. Starting from four fundamental axioms, it defines the speed of light c as the intrinsic interface velocity at which higher-dimensional energy descends into three-dimensional space. It argues that mass is not an intrinsic static property of particles, but a steady-state manifestation of descending energy whose intrinsic velocity is attenuated by crowded collisions and interaction drag in three-dimensional space. This interpretation provides a continuous dynamical account for the mass-energy equivalence relation $E = mc^2$.

Within this framework, the Big Bang singularity corresponds to a white hole channel ejecting energy from four-dimensional into three-dimensional space. In the minuscule volume of the early universe, energy density exceeded the saturation threshold of three-dimensional space. Highly crowded energy units collided frequently, their descent velocity dropped sharply, and matter formed in bulk within an extremely short period—this corresponds to the cosmic inflation epoch. Freely descending energy that did not undergo collision drag maintains propagation at the speed of light and constitutes the cosmic microwave background radiation. Gravity, electromagnetic force and nuclear force are all interpreted as derivative effects of transdimensional tension across different dimensional interfaces at different scales. The apparent randomness of quantum phenomena arises from time-domain undersampling aliasing caused by the bandwidth limit of three-dimensional observation.

Quantum entanglement corresponds to correlated projections of a single higher-dimensional entity at multiple locations in three-dimensional space, which naturally circumvents the paradoxes of local hidden variables and superluminal causality.

This paper also presents multiple testable observational predictions and a complete set of first-order phenomenological dynamical equations, providing directions for subsequent quantitative verification and discriminative experiments. This program constitutes a unified research framework at the ontological level. It does not negate the computational validity of existing physical theories within their domains of applicability, and aims to connect microscopic, macroscopic and cosmic-scale physical pictures with fewer ad hoc assumptions.

Keywords: transdimensional dynamics; cosmic origin; mass-energy conversion; quantum entanglement; dark matter effect; dimensional interface tension

1 Introduction

1.1 Research Background and Open Problems

Over the past half century, cosmology and particle physics have made milestone advances. The ΛCDM cosmological model has accurately fitted the evolution of cosmic large-scale structure via high-precision Planck satellite observations of the cosmic microwave background [1]. The Standard Model of particle physics has successfully unified the electromagnetic, strong and weak interactions, and all particles it predicts have been experimentally verified [2].

Nevertheless, the existing theoretical system still suffers from deep structural limitations:

1. **Theoretical fragmentation**: General relativity describes macroscopic gravity while quantum mechanics describes microscopic particles. The two are fundamentally incompatible at the Planck scale and around black hole singularities, and no consensus solution to the quantum gravity problem has been reached to date [3].

2. **Redundant ad hoc assumptions**: To match observational data, the ΛCDM model introduces multiple independent physical entities including dark matter, dark energy and cosmic inflation, whose nature and microscopic origins remain unknown, giving the model an overall "patchwork" character.

3. **Missing underlying mechanisms**: Special relativity establishes the mass-energy relation $E = mc^2$ but does not explain the dynamical process of conversion between mass and energy. The measurement problem and nonlocality of entanglement in quantum mechanics have persisted with multiple coexisting interpretations, lacking a unified ontological picture [4].

Some frontier quantum gravity approaches, represented by string theory, pursue mathematical self-consistency and symmetry as core principles, introducing elaborate ad hoc structures such as extra dimensions, supersymmetry and compactification. However, they have yet to produce exclusive predictions testable by existing experimental and observational methods, and their physical reality remains widely debated in the academic community.

1.2 Approach of This Work

Instead of the conventional route of appending correction terms to the standard model, this paper attempts to construct a minimal-axiom underlying dynamical framework. Centered on the mechanisms of transdimensional migration, descent and drag of higher-dimensional energy, it integrates gravity, microscopic interactions, mass-energy conversion, quantum phenomena, cosmic origin and dark matter effects into a single ontological picture.

This work is positioned as a unified research program, not a fully verified mature physical theory. Its core value lies in resolving multiple logical fractures in the existing theoretical system with the fewest independent assumptions, and providing a unified underlying picture for subsequent quantitative modeling and experimental testing.

2 Fundamental Axioms and Core Phenomenological Dynamics

All derivations in this program are built upon four basic axioms, with no additional independent ad hoc assumptions.

2.1 Basic Axioms

Axiom 1 — Temporal positivity and entropy increase principle Time is positively irreversible, and entropy increase is the fundamental constraint governing energy and matter conversion. Evolution patterns violating temporal positivity and entropy increase cannot stably exist in this universe.

Axiom 2 — Layered dimensions and transdimensional migration The universe possesses a layered dimensional structure. Higher-dimensional space is not bound by the work constraints of three-dimensional space. Energy can migrate and descend across dimensions. Descent from higher to lower dimensions has an intrinsic interface velocity. The formation and maintenance of dimensional channels require corresponding energy coupling conditions.

Axiom 3 — Energy saturation threshold of three-dimensional space Three-dimensional space has a saturation capacity limit n_{sat} per unit volume for carrying descending higher-dimensional energy. When the number density of energy units exceeds this threshold, energy units become crowded, collide and retard one another.

Axiom 4 — Universal energy conservation The total energy of the entire universe is conserved. Energy is neither created nor destroyed out of nothing; it only transforms between different forms and different dimensional coupling states.

2.2 Definition of Core Physical Quantities and Descent Dynamics Equations

Definition of intrinsic descent velocity v_d: the inherent interface speed at which primordial higher-dimensional energy descends into three-dimensional space. In the ab-

sence of any three-dimensional drag, the intrinsic descent velocity reaches its maximum value c, the speed of light in vacuum.

The total energy of a single energy unit is decomposed into a descent kinetic component and a rest energy component, satisfying global conservation:

$$E = E_k + E_m = \text{constant} \tag{1}$$

where the descent kinetic component takes the first-order phenomenological form $E_k = \frac{1}{2} m_0 v_d^2$, and the rest energy component is $E_m = m(v_d)c^2$. Combining with energy conservation yields the phenomenological relation between rest mass and descent velocity:

$$m(v_d) = m_0 \left(1 - \frac{v_d^2}{2c^2}\right) \tag{2}$$

When $v_d = c$, rest mass approaches zero, corresponding to pure radiation states such as photons. When $v_d \ll c$, it reduces to the steady rest mass m_0, corresponding to ordinary matter.

The evolution of descent velocity under three-dimensional collision drag satisfies the first-order dynamical equation:

$$\frac{dv_d}{dt} = -\alpha \cdot n \cdot v_d \tag{3}$$

where α is the three-dimensional collision drag coefficient (a phenomenological parameter to be fitted) and n is the local number density of energy units.

3 Cosmic Origin and Early Evolution: White Hole Singularity and Inflation Mechanism

3.1 The Ontology of the Big Bang: White Hole Energy Eruption

This program proposes that the Big Bang singularity is essentially a white hole channel injecting primordial four-dimensional energy into three-dimensional space. At the birth of the universe, the white hole instantaneously injected high-flux higher-dimensional descending energy into an extremely small three-dimensional volume, with initial energy density far exceeding the three-dimensional saturation threshold n_{sat}.

3.2 Bulk Matter Generation and Cosmic Inflation

According to Axiom 3, when energy density exceeds the saturation threshold, energy units become highly dense and collide frequently. The intrinsic descent velocity drops drastically within a very short time, corresponding to the bulk generation of rest mass. The generation rate of matter number density follows the phenomenological equation:

$$\frac{dn_m}{dt} = k \cdot n^2 \cdot \left(1 - \frac{n_{\text{sat}}}{n}\right) \cdot H(t)^{-1} \tag{4}$$

where k is the collision solidification efficiency coefficient and $H(t)$ is the Hubble parameter describing the dilution effect of spatial expansion. When $n \gg n_{\text{sat}}$, the generation

rate is proportional to the square of density, corresponding to inflationary matter production in the high-density early universe. When $n \to n_{\text{sat}}$, the generation rate approaches zero and the universe enters the structure-formation stage dominated by gravity.

This mechanism directly explains why cosmic matter formed predominantly in the very early epoch and why the modern universe no longer produces matter on large scales: spatial expansion has diluted the energy density such that the crowding-collision solidification condition no longer holds.

3.3 Origin of the Cosmic Microwave Background Radiation

Not all energy erupted from the white hole participates in crowding collisions and mass generation. A fraction of energy units undergo no effective collision coupling after ejection; their intrinsic descent velocity remains near c, unattenuated, unsolidified and free of rest mass. This freely propagating primordial descending energy is what we observe today as the cosmic microwave background (CMB) radiation.

This mechanism answers a classic cosmological question—why the relic radiation from the Big Bang still propagates at near-light speed after billions of years without converting into matter: it is essentially unretarded pure descending energy that does not satisfy the collision-coupling condition for mass generation.

4 Unified Interpretation of Fundamental Interactions: Homologous Transdimensional Tension Effects

This program holds that the four fundamental interactions are not independent elementary forces, but derivative manifestations of transdimensional tension between dimensional interfaces at different dimensional combinations and different spatial scales.

4.1 Gravity: 3D-to-2D Transdimensional Descent Tension

Gravity is the spatial distortion effect caused by three-dimensional energy descending into two-dimensional space. Let ε denote the spatial deformation variable; it satisfies a Poisson-type phenomenological equation with the local rest mass density ρ_m:

$$\nabla^2 \varepsilon = 4\pi G_{\perp} \rho_m \tag{5}$$

where $G_{\perp}$ is the transdimensional tension coefficient. Under the weak-field, low-velocity, small-deformation approximation, this equation reduces to Newton's gravitational Poisson equation and is fully compatible with everyday observations. The greater the total solidified matter in a unit volume, the larger the distortion amplitude of transdimensional descent and the stronger the gravitational effect.

4.2 Electromagnetic Force: 4D-to-3D Transdimensional Transition Tension

Electrons are incompletely frozen projections of higher-dimensional energy. They inherently tend to transition back into four dimensions, generating reverse transdimensional

attraction that corresponds to the attraction between opposite charges. Meanwhile, constrained by the three-dimensional energy saturation threshold, accumulation of like high-energy projections produces saturation repulsion, corresponding to the repulsion between like charges.

4.3 Nuclear Force: Short-Range Strong Coupling Tension at the 4D–3D Interface

The nuclear force arises from strong coupling binding tension at the 4D–3D interface at extremely short distances. Its distance dependence is described by a Yukawa-like potential:

$$V(r) = -V_0 \cdot \frac{e^{-r/r_0}}{r/r_0} \tag{6}$$

where r_0 is the critical distance of transdimensional coupling (on the order of 10^{-15} m) and V_0 is the coupling potential depth. When particle separation is below the critical scale, the transdimensional projection interfaces fit tightly, coupling is established, and strong binding emerges. When separation exceeds the critical value, transdimensional coupling decays exponentially and the force vanishes rapidly, perfectly matching the experimental features of the nuclear force: short range, strong attraction and saturation.

4.4 Black Holes, White Holes and Dark Matter Effects

- **Black hole**: A stable channel for three-dimensional energy to descend into two dimensions, sustained by continuous matter inflow. When matter supply is insufficient, the dimensional channel gradually dissipates.
- **White hole**: A channel injecting four-dimensional energy into three dimensions. The Big Bang singularity is a cosmic-scale white hole event. Black holes and white holes are distinct types of dimensional channels; there is no topological "black-hole-to-white-hole" connection.
- **Dark matter effect**: Originates from residual 3D–2D transdimensional distortion tension left behind by extinct ancient black holes. The residual gravitational potential obeys the following attenuation relation with time and distance:

  $$\Phi(r,t) = \Phi_0 \cdot \frac{e^{-t/\tau}}{r} \cdot e^{-r/r_d} \tag{7}$$

  where τ is the transdimensional tension relaxation time and r_d is the spatial attenuation scale of the tension. Dark matter signals concentrate in regions that once hosted black holes; cosmic voids have weak residual tension and low dark matter abundance.

The weak interaction is preliminarily identified as a metastable coupling dissociation effect at the dimensional interface; detailed dynamical description remains to be completed in future work.

5 Transdimensional Projection Interpretation of Quantum Phenomena

5.1 Origin of Apparent Randomness: Time-Domain Undersampling Aliasing

The full ontology of a quantum object resides in higher-dimensional space, and its state evolution is a deterministic, continuous temporal process with no intrinsic randomness.

All observational means in the three-dimensional world, bounded by three-dimensional causality and signal speed, have an inherent minimum time slice—an upper limit on the observation sampling frequency f_{obs}. The intrinsic evolution frequency f_d of the higher-dimensional ontology is far higher than the three-dimensional sampling frequency, satisfying the Nyquist undersampling condition:

$$f_{\text{obs}} < 2f_d \tag{8}$$

Under this condition, observation suffers from time-domain aliasing: the sampling phase is misaligned with the higher-dimensional evolution phase, and measurement results exhibit pseudo-random behavior. The quantum uncertainty principle is essentially a sampling aliasing effect caused by limited observation bandwidth, not an inherent indeterminacy of the object itself.

Let ω_d be the angular frequency of higher-dimensional ontological evolution and ω_s the angular frequency of observation sampling. The apparent aliasing frequency after sampling is:

$$\omega_{\text{app}} = |\omega_d - k\omega_s| \tag{9}$$

where k is the integer satisfying the aliasing condition. The apparent probability distribution of measurement results is determined by the phase average of the deterministic higher-dimensional trajectory over discrete sampling points:

$$P(x) = \frac{1}{T_s} \int_0^{T_s} \delta\left(x - x_d(t)\right) dt \tag{10}$$

where T_s is the sampling period and $x_d(t)$ is the deterministic evolution trajectory of the higher-dimensional ontology.

5.2 Quantum Entanglement Mechanism: Correlated Projections of a Single Higher-Dimensional Entity

A core postulate of this program: there are no two separate entangled particles. The observed phenomenon of quantum entanglement corresponds to multiple spatially separated projections of one and the same higher-dimensional entity onto three-dimensional spacetime. Two projections that appear distant from a three-dimensional perspective share the same higher-dimensional ontological evolution sequence, so their states are inherently correlated. No superluminal signal needs to propagate through three-dimensional space, and thus three-dimensional light-speed causality is not violated.

This mechanism naturally evades the local hidden-variable constraint of Bell's inequality. The derivation of Bell's inequality presupposes that "two subsystems are local independent entities"; in this framework, the two projections are not independent entities, so the premise does not apply, and there is no contradiction with experimental observations of Bell inequality violation [5].

5.3 Verification Scheme

Precisely record the timestamp, measurement basis and measurement outcome for every observation in an entanglement experiment. Perform spectral analysis on the time-series dataset to search for a reproducible intrinsic frequency of higher-dimensional evolution that can predict subsequent measurement results. The existence of such a stable frequency would provide direct experimental support for this model.

6 Observable Predictions and Discriminative Tests

If this program could only explain known phenomena, it would remain a physical interpretation. To qualify as a testable physical theory, it must make unique predictions distinguishable from the standard model. This paper proposes three observable discriminative directions:

6.1 Prediction 1: Periodic Modulation in High-Time-Resolution Quantum Measurements

Since quantum randomness originates from observation undersampling aliasing, as the time resolution of measurement improves and approaches the limit of the three-dimensional minimum time slice, the quantum statistical distribution will exhibit identifiable periodic modulation. In standard quantum mechanics, probability rules are independent of observation time resolution and no such effect exists. Detection of this modulation would provide direct experimental evidence for the framework.

6.2 Prediction 2: Spatial Correlation Between Dark Matter Distribution and Ancient Black Hole Remnants

Because dark matter effects arise from transdimensional tension left by ancient black holes, dark matter density peaks should be positively correlated with regions of historical black hole formation. In large cosmic voids, dark matter abundance should be significantly lower than the uniform distribution predicted by the ΛCDM model. This prediction can be tested with next-generation weak gravitational lensing surveys and galaxy survey data.

6.3 Prediction 3: Systematic Deviations in Early-Universe Element Abundance Fitting

Based on the crowding-drag matter-formation model, quantitative fits to primordial nucleosynthesis element abundances will show distinguishable systematic deviations from the standard ΛCDM model. This prediction can be verified via high-precision observations of primordial element abundances.

7 Discussion and Outlook

7.1 Strengths of the Framework

1. **Minimal axioms, unified narrative**: With only four basic axioms, it connects cosmic origin, mass-energy conversion, fundamental interactions, quantum phenomena and dark matter effects, without introducing numerous independent ad hoc assumptions.

2. **Resolution of classical paradoxes**: It explains the mass-energy conversion process at the underlying dynamical level, and resolves the tension between entanglement nonlocality and light-speed causality via the projection mechanism, rather than setting them aside by demarcating domains of applicability.

3. **Compatibility with existing physics**: It does not deny the computational accuracy of relativity, quantum mechanics and ΛCDM within their respective domains, and regards them as effective approximations of this framework under specific conditions.

7.2 Limitations and Future Work

This study currently stands at the stage of an ontological program. A complete and unified system of quantitative dynamical equations has not yet been established, and much work remains before it can be considered a mature physical theory:

1. Construct a unified transdimensional geometric-temporal differential equation system for a unified mathematical description of all four fundamental interactions.

2. Rigorously distinguish intrinsic descent velocity from three-dimensional macroscopic velocity, and establish transformation rules compatible with experimentally verified Lorentz transformations.

3. Refine the quantitative model of early cosmic inflation and element abundances, and perform fitting comparisons with CMB observational data.

4. Design high-time-resolution quantum experiments to test the periodic modulation prediction.

7.3 Boundary in Philosophy of Science

Even if quantitative modeling is completed and some observational tests are passed in the future, experimental data alone cannot ”prove” that the higher-dimensional descent picture is the objective truth of the universe. The problem of theoretical equivalence in philosophy of science holds that multiple fundamentally different underlying models can yield identical observational predictions. The core role of experiment is falsification, not exclusive confirmation of any single ontological picture. The core value of this program is to provide a unified cosmic picture with fewer contradictions, fewer assumptions and greater coherence, offering an alternative path for fundamental physics research distinct from the mainstream paradigm.

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